

Loops

Homework: for, while & Repeating Yourself

Name:

Date:

No computer needed! Show what you remember about loops. Finish by designing a loop of your own.

Part 1 — Matching

Write the letter of the correct meaning in the box next to each piece of code.

- A. Repeats a set number of times, with counting built in.
- B. Repeats as long as something stays true.
- C. A shortcut that means 'add 1'.
- D. A shortcut that means 'subtract 1'.
- E. A variable made outside a loop to add things up.
- F. A loop that never stops because the check is always true.

for

while

i++

i--

accumulator

infinite loop

Part 2 — True or False

Write T (true) or F (false) in each box.

A for loop has three parts separated by semicolons.

The loop checks its condition before each round.

If you forget to change the counter, the loop may run forever.

An accumulator should be created inside the loop.

An infinite loop can freeze the page you are running it on.

Part 3 — Predict the Output

Write exactly what each loop would print.

Code:

```
for (let i = 1; i <= 3; i++) {  
  console.log(i)  
}
```

Prints:

Code:

```
for (let i = 3; i >= 1; i--) {  
  console.log(i)  
}
```

Prints:

Code:

```
for (let i = 0; i <= 12; i = i + 4) {  
  console.log(i)  
}
```

Prints:

Code:

```
let total: number = 0  
for (let i = 1; i <= 4; i++) {  
  total = total + i  
}  
console.log(total)
```

Prints:

Part 4 — Spot the Error

Each loop has ONE problem. Describe it and write the fix.

Snippet 1:

```
let x: number = 5  
while (x > 0) {  
  console.log(x)  
}
```

What's wrong, and how do you fix it?

Snippet 2:

```
for (let i = 1; i <= 5; i++) {  
  let total: number = 0  
  total = total + i  
}
```

What's wrong, and how do you fix it?

Part 5 — Short Answer

1. Explain in your own words what the three parts of a for loop do.

2. When would you use a while loop instead of a for loop?

Part 6 — Design a Loop

Invent a small looping program on paper.

What does your loop do?

Write your code here (it's okay if it isn't perfect!):